UpToDate®
Official reprint from UpToDate®
www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Initial management of the critically ill adult with an unknown overdose

AUTHOR: Marco L A Sivilotti, MD, MSc, FRCPC, FACMT, FAACT
SECTION EDITOR: Evan Schwarz, MD
DEPUTY EDITOR: Michael Ganetsky, MD

All topics are updated as new evidence becomes available and our peer review process is complete.

Literature review current through: **Jan 2026.**
This topic last updated: **Dec 04, 2025.**

INTRODUCTION

Poisoning is a leading cause of death especially in the young, in whom it is the leading cause of non-traumatic cardiac arrest under the age of 35 years. Overdose, both intentional and unintentional, has also become the leading cause of injury-related death in the United States, exceeding the number of deaths due to firearms, falls, or motorized vehicle collisions [1,2].

Severely poisoned patients may present *in extremis*. Such patients require an organized, targeted resuscitation despite incomplete, uncertain, or even erroneous information. A "generic" approach based upon the advanced cardiac life support (ACLS) protocols intended for cardiac patients is suboptimal [3,4]. It can lead to missed opportunities for specific life-saving interventions and may at times be harmful.

This topic will describe an approach to the resuscitation of the critically ill poisoned adult patient when the identity of the causative agent(s) is initially unknown. The general approach to the poisoned patient and the management of specific poisonings are described separately. (See "General approach to drug poisoning in adults".)

INITIAL DATA ACQUISITION

Any readily available information about the patient and the poisoning should be obtained from pre-hospital care providers, other first responders (including witnesses, firemen, police, friends, and family), and from medical records. Medication or chemical product containers, material safety data sheets, pharmacy records, and institutional or patient lists of prescribed medication can be helpful. In addition, the setting and circumstances may help identify the most likely toxin(s) involved and suggest specific interventions. Some scenarios, such as recreational ingestion of illicit alcohol (eg, containing methanol), inhalation of toxic gases in an enclosed space, or accidental industrial releases/fires suggest the need to prepare for multiple victims. A person who rapidly decompensates after being taken into police custody may have "stuffed" or "packed" large amounts of illicit drugs requiring the immediate removal of the leaking package from a body cavity [5]. (See "General approach to drug poisoning in adults", section on 'History' and "Acute ingestion of illicit drugs (body stuffing)" and "Internal concealment of drugs of abuse (body packing)".)

FIRST PRIORITIES

In addition to securing the airway, breathing, and circulation (ABC's) as with any critically ill or injured patient, the resuscitation leader must consider two additional imperatives that may arise with severely poisoned patients: preserving the operational capacity of the emergency health care system and ensuring the safety of health care workers. These priorities are addressed concurrently as part of the primary assessment.

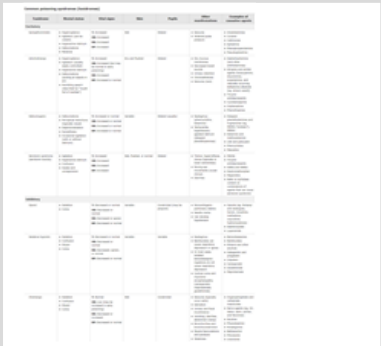
Multiple casualty and mass casualty incidents may involve toxic exposures and challenge health care systems of any size. Even a single patient contaminated or potentially contaminated with a highly potent toxin (eg, a radioactive isotope or nerve agent such as sarin) can disable an entire emergency department (ED).

Highly potent toxins can directly threaten the health of clinicians and rescue personnel when proper protection is not used. However, such agents are extremely rare outside of a large-scale industrial accident, major transportation accident, or terrorist attack. Opioid toxicity is extremely unlikely following incidental dermal contact with fentanyl or fentanyl analogs, despite unfounded concerns for first responders rendering first aid and resuscitation to a patient with opioid intoxication [6,7]. Routine measures are sufficient for nearly all single-patient encounters involving consumer products and common chemicals. If readily available, nitrile gloves, safety goggles, and a water-resistant gown can be worn for additional security when dealing with extensive solid particle contamination, liquid spills, or emesis following an unknown exposure. However, locating such protective equipment should not delay the delivery of lifesaving care without clear reason to suspect a highly toxic agent [8].

Ideally, surface decontamination should be performed at or near the scene (ie, prior to transport) or before the patient is brought into the clean area of the ED (if not performed in the pre-hospital setting). Clothing soiled with chemicals or bodily fluids should be placed immediately into appropriate closed containers and disposed of properly. Decontamination procedures and the chemical and biologic agents likely to be involved in an exposure are discussed separately. (See "Topical chemical burns: Initial evaluation and management" and "Chemical terrorism: Rapid recognition and initial medical management" and "Detection of and response to biologic terrorism".)

RAPID FIRST LOOK: EXAMINATION, MONITORING, AND TESTING

A rapid systematic examination should be performed while resuscitative measures are initiated. The class of toxin involved may be suggested by characteristic combinations of symptoms and signs (so-called "toxidromes"). The "toxidrome-oriented" physical examination consists of vital signs; level of alertness; pupil size and position; mucous membrane moisture and secretions; skin temperature and moisture; presence or absence of bowel sounds; and motor tone. Alertness can be categorized simply as alert, responsive to voice, responsive to pain, or unresponsive, which is preferred over the Glasgow Coma Scale (GCS) to characterize the mental status of a poisoned patient. The findings of this rapid initial examination may provide valuable insight into the class of toxin involved. The major toxidromes and their associated findings are summarized in the accompanying table (table 1). Disturbances in temperature (table 2), heart rate and blood pressure (table 3), and respiration (table 4) can also suggest a toxin class.



Common poisoning syndromes (toxidromes)

Table 1 - larger image below

In addition to the findings above, the clinician should look for signs of trauma (self-inflicted or accidental), recent parenteral drug use, transdermal patches, and seizures (eg, incontinence, lateral tongue laceration). Cervical spinal motion restrictions may be needed if major trauma is suspected.

The clinical team should immediately obtain intravenous access, apply a cardiac monitor and pulse oximeter, and provide high flow oxygen by non-rebreather face mask, as needed. The following laboratory tests should usually be obtained:

- Point-of-care capillary blood glucose
- Complete blood count
- Basic serum electrolytes, BUN, and creatinine
- Serum lactate
- Venous or arterial blood gas, including carboxyhemoglobin and methemoglobin level in appropriate circumstances (eg, smoke inhalation or cyanosis not improving with supplemental oxygen, respectively)
- Serum acetaminophen and salicylate concentrations following self-harm
- Electrocardiogram
- Other specific drug concentrations as indicated (eg, ethanol, theophylline, digoxin, anti-epileptics)

An initial urine sample can be retained for comprehensive drug testing in a critically-ill patient with an unclear etiology, a prolonged clinical course, or an unusual clinical course (eg, multiple substances creating mixed toxicity). Unlike the commonly available urine "drug of abuse screen," which is an immunoassay panel that rarely alters the initial management of most patients, comprehensive drug testing requires specialized expertise and equipment, is not available in most hospital laboratories and has long turnaround times. The first available urine sample is generally preferred, ideally obtained from the initial catheterization without intraurethral lidocaine, which may mask the presence of other substances. (See "Urine drug testing".)

More detailed information about these initial tests is provided separately. (See "General approach to drug poisoning in adults".)

SYSTEMATIC EVALUATION: THE "ABCDE" APPROACH

We have adapted the basic "ABC" (airway, breathing, circulation) resuscitation approach to the initial management of the critically ill adult with an unknown overdose. The steps are organized sequentially as a mnemonic according to the issues that pose the most immediate life threats. Treatment for a given problem can be initiated immediately in the order encountered, but multiple treatments may need to be pursued concurrently in complex resuscitations. Important modifications to routine resuscitation specific to the poisoned patient are described in each section below.

"A": Airway stabilization — Patients who cannot protect their airway should be tracheally intubated immediately. The evaluation of the patency or protection of the airway is discussed separately. (See "The decision to intubate", section on 'Is patency or protection of the airway at risk?'.)

Exceptions include suspected opioid overdose and severe hypoglycemia. If opioid toxicity is suspected (table 1), administer naloxone while assuring adequate oxygenation and ventilation [9]. Use small doses initially (eg, 0.04 or 0.05 mg intravenously or 0.1 mg intramuscularly) when opioid dependence is possible and ventilation can be maintained, and doubling the prior dose until reversal of respiratory depression is achieved. Severe hypoglycemia should also be ruled out (with a point-of-care capillary blood glucose) as a cause of depressed mental status prior to intubation. (See "Acute opioid intoxication in adults", section on 'Basic measures and antidotal therapy'.)

Do **not** routinely administer flumazenil empirically for possible benzodiazepine overdose, as the risk benefit ratio of this agent is poor, and withdrawal seizures can develop [9]. The use of flumazenil (and scenarios where it may be appropriate) is discussed separately. (See "Benzodiazepine poisoning", section on 'Role of antidote (flumazenil)'.)

Placing a nasopharyngeal airway (ie, nasal trumpet) may be sufficient to overcome upper airway obstruction from central nervous system depression (CNS) and to preclude the need for tracheal intubation in a patient who has ingested a short-acting sedative (eg, ethanol, gamma-hydroxybutyrate) and whose clinical course is likely to improve. (See "Basic airway management in adults", section on 'Nasopharyngeal airway'.)

We do not solely use a Glasgow Coma Scale (GCS) (table 5) ≤8 to identify the need for tracheal intubation in an unresponsive poisoned patient who is protecting their airway, maintaining adequate oxygen saturation, hemodynamically stable, and not expected to deteriorate based on the suspected ingestion (eg, short-acting sedative such as ethanol). The GCS was developed to classify injury severity and prognosticate in patients with traumatic brain injury (TBI); a GCS ≤8 represents severe TBI and is generally used as an indication for tracheal intubation. Unlike comatose patients with TBI, poisoned patients generally have a much better prognosis despite a low GCS [10]. The prognosis is particularly favorable in patients intoxicated with short-acting sedatives (eg, ethanol) since they typically present for care near or past the time of maximal CNS depression and improve rapidly if they have a patent airway and spontaneous respiration. For these patients, the risks of tracheal intubation (eg, hemodynamic instability, hypoxia, dental injury, barotrauma, ventilator-associated pneumonia) are unlikely to outweigh the expected benefit (ie, aspiration prevention). Evidence includes a multicenter trial of 225 adult patients with acute poisoning and a GCS ≤8 randomized to have selective tracheal intubation based on respiratory distress (ie, pulse oximetry <90 percent despite nasal canula oxygen), seizure, vomiting, or shock (ie, systolic blood pressure <90 mmHg after 1 liter of crystalloid fluid) versus routine care with permissive intubation at the discretion of the treating physician [11]. Patients in the selective arm were less likely to undergo mechanical ventilation (18.1 versus 59.6 percent, absolute difference -42.5, 95% CI -30.9 to -54.1), less likely to be admitted to the intensive care unit (39.7 versus 66.1 percent, absolute difference -29.2, 95% CI -17.4 to -41), experienced fewer adverse events from intubation (6 versus 14.7 percent,

absolute difference -8.6, 95% CI -0.7 to -16.6), less likely to develop pneumonia (5.4 versus 15 percent, absolute difference -9.6, 95% CI -17.5 to -1.7), and had a shorter hospital length of stay (median 21.5 versus 37 hours, relative risk 0.74, 95% CI 0.53-1.03); the latter finding did not achieve statistical significance. Importantly, the trial excluded patients who required immediate tracheal intubation (signs of respiratory distress, clinical suspicion of any brain injury, seizure, shock) or had cardiotoxic drug overdoses. Also, patients were young (mean age 33 years), there were no deaths, most patients ingested sedatives (ethanol in 67 percent), and almost all exposures involved substances that would be expected to have a benign course.

Unless the patient is moribund, rapid sequence intubation with pre-oxygenation and neuromuscular blockade is typically the best approach to securing the airway [12]. Rocuronium is preferred over succinylcholine in cases of suspected organophosphate poisoning and acute digoxin toxicity. In organophosphate poisoning, succinylcholine may have a markedly prolonged duration of action, as the cholinesterases which degrade it are inactivated by the toxin. Digoxin poisoning may produce hyperkalemia, a contraindication to succinylcholine use. (See "Rapid sequence intubation in adults for emergency medicine and critical care" and "Organophosphate and carbamate poisoning", section on 'All patients: Supportive care' and "Digitalis (cardiac glycoside) poisoning", section on 'Management'.)

Patients with a caustic ingestion who manifest stridor and drooling are ideally tracheally intubated early by the most experienced clinician available using an awake approach. An alternate strategy for securing the airway, and the means to establish a surgical airway if necessary, must be immediately available. (See "Approach to the anatomically difficult airway in adults for emergency medicine and critical care", section on 'Awake techniques'.)

"B": Breathing — Administer supplemental oxygen as needed to all critically ill patients with a suspected overdose. While useful in mechanically ventilated patients, the benefit of end tidal carbon dioxide monitoring (ie, capnography) in spontaneously breathing overdose patients is less certain, especially when such patients are routinely administered high doses of oxygen. Capnometry should not be relied upon to monitor the adequacy of ventilation in deeply sedated patients, especially since the administration of supplemental oxygen renders pulse oximetry less effective as an early warning device for hypoventilation. (See "Carbon dioxide monitoring (capnography)".)

Several toxins interfere directly with tissue oxygenation. Carbon monoxide poisoning can cause severe hypoxia despite falsely normal pulse oximetry readings. Methemoglobinemia can cause the pulse oximeter to falsely read an oxygen saturation around 85 percent despite visible cyanosis. Toxicologic causes of cellular hypoxia despite normal oxygen saturation and delivery include cyanide, hydrogen sulfide, and sodium azide. Patients with profound tissue hypoxia from these toxins require endotracheal intubation and mechanical ventilation using 100 percent oxygen. (See "Carbon monoxide poisoning" and "Methemoglobinemia" and "Cyanide poisoning".)

Toxins can also directly affect ventilation. Patients with aspirin poisoning are initially tachypneic and hyperpneic due to the stimulatory effects of aspirin upon the respiratory center of the medulla. This hyperventilation is beneficial, and intubation should be avoided if possible because it is difficult to replicate such high minute ventilation with mechanical ventilation. Nevertheless, patients unable to protect their airway must be intubated and hyperventilated. Severe aspirin poisoning can cause precipitous decompensation and death shortly after intubation, and such patients require specific and aggressive resuscitation, including treatment with sodium bicarbonate, high minute ventilation, and immediate hemodialysis. (See "Salicylate (aspirin) poisoning: Management", section on 'ABCs and supportive care'.)

Patients with profound metabolic acidosis due to other toxins (eg, methanol, ethylene glycol) can also develop extremely high compensatory minute ventilation. These patients may be acidemic despite a $PaCO_2$ near 10 mmHg. When this high minute ventilation rate slows (eg, due to seizures, fatigue, or iatrogenic sedation or paralysis), the arterial pH falls abruptly, sometimes well below 7. As such, patients with severe metabolic acidosis due to poisoning are at risk of decompensating rapidly from respiratory failure. Such patients should be intubated for any sign of impending respiratory failure, either clinically or on blood gas testing. A seemingly normal $PaCO_2$ of 30 to 40 mmHg in the face of a severe metabolic acidosis is evidence of respiratory failure. (See "Methanol and ethylene glycol poisoning: Management", section on 'Overview of management'.)

When intubating hyperpneic patients with severe metabolic acidosis, we suggest the following approach:

- Administer one to three 50 mL amps of 7.5 to 8.4 percent sodium bicarbonate via intravenous bolus before and after intubation.

- Avoid sedation until just prior to tracheal intubation.
- Optimize the likelihood of first-pass success by using neuromuscular blockade and sound technique; the most experienced intubator available should perform the procedure. (See "Rapid sequence intubation in adults for emergency medicine and critical care" and "Direct laryngoscopy and endotracheal intubation in adults".)
- Maximize minute ventilation once intubated, while following peak airway pressures and arterial blood gases. Initial settings could be 12 mL/kg at 20 breaths per minute for an adult, with frequent reassessment of acidemia and for signs of barotrauma. As the acidosis resolves, minute ventilation should be reduced. (See "Mechanical ventilation of adults in the emergency department".)
- Organize immediate hemodialysis if a dialyzable toxin (eg, aspirin, methanol, ethylene glycol) is responsible.

"C": Circulation

Asystole and ventricular fibrillation — Asystole and ventricular fibrillation are managed according to standard ACLS protocols (algorithm 1). (See "Advanced cardiac life support (ACLS) in adults", section on 'Pulseless ventricular tachycardia and ventricular fibrillation' and "Advanced cardiac life support (ACLS) in adults", section on 'Asystole and pulseless electrical activity'.)

Hypotension — Hypotension is treated with a rapid intravenous (IV) infusion of up to 2 L of isotonic crystalloid, followed by a norepinephrine infusion or small boluses of phenylephrine, pending more specific information (eg, bedside echocardiography, point-of-care ultrasonography) to guide therapy. (See "Evaluation of and initial approach to the adult patient with undifferentiated hypotension and shock", section on 'Point-of-care ultrasonography'.)

At times, the dose of vasopressor or inotrope needed to achieve a response may be much higher than doses used for non-poisoned patients [13]. Circulatory assist devices (eg, intraaortic balloon pump) and extracorporeal life support (eg, circulatory bypass pump or veno-arterial extracorporeal membrane oxygenation) may be needed in refractory shock [14,15]. Ingestions that have been successfully managed with extracorporeal life support are listed in the table (table 6). (See "Extracorporeal life support in adults in the intensive care unit: Overview".)

Selected antidotes can be administered empirically to hypotensive patients when circumstances suggest a specific etiology. A list of these antidotes with empiric dosing guidelines is provided (table 7).

Despite numerous case reports of its use for refractory shock, there is scant evidence of benefit for lipid emulsion therapy when used for drug toxicity (other than local anesthetics), and its routine use is not recommended [16,17]. Methylene blue has also been proposed for empirical use to treat refractory shock in overdose based on a similarly limited evidence base [18]. (See "Calcium channel blocker poisoning", section on 'Lipid emulsion therapy' and "Local anesthetic systemic toxicity" and "Methemoglobinemia", section on 'Methylene blue (MB)'.)

Empirical dosing of emergency antidotes to adult patients in extremis*

Suspected toxin (or similar agent)	Antidote
Cyanide	5 g hydroxocobalamin IV, repeat x 1 (alternative is nitrite/thiosulfate kit)
Methemoglobinemia	2 mL/kg 2 percent methylene blue IV over 5 minutes
Digoxin	2 to 5 vials digoxin antibody (Fab) fragments IV
Calcium channel blocker	20 mL 10 percent calcium chloride IV, repeat x 2; 0.5 to 1 units/kg/hour insulin (and dextrose) IV
Local anesthetic	1.5 mL/kg of 20 percent intralipid bolus IV, may repeat several times (alternatively, after bolus infuse 0.25 mL/kg per minute for 30 minutes; maximum dose 10 mL/kg)
Sodium channel blockade	100 mL of 7.5 percent to 8.4 percent sodium bicarbonate IV, repeat x 2
Cholinergic agent	2 mg atropine IV , doubled every 3 minutes as needed
Isoniazid	5 g pyridoxine over 10 minutes or until seizures stop; remainder over 4 hours
Chloroquine	2 mg/kg diazepam IV over 30 minutes

Empirical dosing of emergency antidotes to adult patients in extremis*

Table 7 - larger image below

Bradycardia with hypotension — Bradycardia with hypotension suggests an overdose with digoxin, calcium channel blockers, or beta blockers. Digoxin toxicity may be characterized by increased automaticity, depressed AV nodal conduction, characteristic repolarization changes, and gastrointestinal symptoms. Beta-blocker toxicity can be accompanied by hypoglycemia and, in the case of propranolol, sodium channel blockade with widening of the QRS complex and seizures. Overdose with calcium channel blockers usually causes hyperglycemia and a relatively well-preserved mental status despite hypotension. Specific therapy for each agent is available and discussed separately. (See "Digitalis (cardiac glycoside) poisoning" and "Dosing regimen for digoxin-specific antibody (Fab) fragments in patients with cardiac glycoside (digoxin) toxicity" and "Calcium channel blocker poisoning", section on 'Management' and "Beta blocker poisoning", section on 'Management'.)

Monomorphic, wide-complex tachycardia — Monomorphic, wide-complex tachycardia in the setting of poisoning is commonly supraventricular in origin with aberrancy, often due to sodium channel blockade. Many poisons can cause

sodium channel blockade, including tricyclic antidepressants, antihistamines, Type IA antiarrhythmics, and cocaine. The treatment is sodium bicarbonate 50 to 100 mEq given as an intravenous (IV) bolus, and repeated as necessary, until the QRS interval is <100 msec in the limb leads, or until arterial pH approaches 7.55. Other Type I antiarrhythmics (eg, procainamide) should be avoided. (See "Tricyclic antidepressant poisoning", section on 'Sodium bicarbonate for cardiac toxicity' and "Anticholinergic poisoning" and "Cocaine: Acute intoxication".)

Polymorphic ventricular tachycardia — Polymorphic ventricular tachycardia (torsade de pointes) is occasionally seen following overdose with Types IA, IC, and III antiarrhythmics; pentamidine; antipsychotics; arsenicals; antifungals; and antihistamines. This arrhythmia is treated with 2 g magnesium sulfate IV given over two to five minutes. Up to two additional doses may be given as needed. Overdrive pacing with isoproterenol or overdrive electrical pacing may also be used, as bradycardia potentiates this arrhythmia [19]. (See "Advanced cardiac life support (ACLS) in adults", section on 'Irregular wide complex'.)

Narrow complex tachycardia — Narrow complex tachycardia and hypertension due to drug overdose or withdrawal are usually due to hyperadrenergic states, as seen with cocaine, amphetamine, and other sympathomimetics. Treatment consists primarily of benzodiazepines, and perhaps phentolamine for severe hypertension [20,21]. Refractory hypertension may also be treated with nitroglycerin (in the setting of cardiac ischemia), clevidipine, nicardipine, or nitroprusside. Beta-blockers and antipsychotics should be avoided [22,23]. Synchronized cardioversion is rarely effective. (See "Cocaine: Acute intoxication", section on 'Initial management' and "Methamphetamine: Acute intoxication", section on 'Management'.)

"D": Disability and neurological stabilization — Once the airway, breathing, and circulation are secured, attention is next directed towards neurologic stabilization. The so-called "coma cocktail" of dextrose, oxygen, naloxone, and thiamine given empirically is an outdated concept and has been replaced by selective use of each component as necessary [9,24]. Hypoglycemia can present as **any** alteration of mental status, including confusion, seizures, focal deficits, and coma. Treat these symptoms with intravenous dextrose when the glucometer reports a low or low-normal capillary blood glucose.

Naloxone is indicated for the reversal of respiratory depression in the setting of known or suspected opioid toxicity (see "Acute opioid intoxication in adults", section on 'Basic measures and antidotal therapy'). Flumazenil should **not** be used empirically, even when benzodiazepine toxicity is suspected, because it can precipitate benzodiazepine withdrawal, causing seizures (see "Benzodiazepine poisoning", section on 'Role of antidote (flumazenil)'). Thiamine may be administered safely before glucose, but is unlikely to reverse coma [25]. Physostigmine should not be used as a non-specific analeptic, but does play a selected role in the management of severe anticholinergic (antimuscarinic) poisoning. (See "Anticholinergic poisoning", section on 'Antidotal therapy with physostigmine for severe toxicity'.)

Seizures due to poisoning or withdrawal are best treated with escalating doses of benzodiazepines (eg, diazepam 5 mg IV, repeated and doubled every 5 to 10 minutes as necessary for refractory seizures). Phenytoin should be avoided as it is usually ineffective, and can cause toxicity due to the propylene glycol solvent. Propofol has superseded the barbiturates and inhalational anesthetics as second-line therapy for benzodiazepine-resistant seizures [26]. When seizures are likely due to sodium channel blockade, intravenous sodium bicarbonate should be administered in addition to benzodiazepines. Dosing is described above (see '"C": Circulation' above). High doses of pyridoxine (5 g IV) should be given to patients in status epilepticus believed to be caused by isoniazid overdose or monomethylhydrazine poisoning (eg, Gyromitra mushrooms). (See "Isoniazid (INH) poisoning".)

Causes other than overdose may be responsible for coma or depressed mental status in the poisoned patient. This is discussed separately. (See "Stupor and coma in adults".)

"E": Exposure and elimination — This phase of management aims to remove clothing, transdermal medication patches, and other external contaminants; to measure core temperature and treat hypothermia or hyperthermia as necessary; to identify self-inflicted or accidental trauma; and to search personal items for concealed drugs, weapons of self-harm, or clues regarding medical history and the nature of the overdose.

Severe hyperthermia (greater than 40°C) usually requires paralysis, sedation, and ice bath cooling (see "Nonexertional (classic) heat stroke in adults", section on 'Management'). Severe hypothermia (less than 30°C) is also treated aggressively with rapid rewarming (see "Accidental hypothermia in adults: Management"). Therapeutic hypothermia should be

administered to unconscious patients with return of spontaneous circulation after circulatory arrest. (See "Initial assessment and management of the adult post-cardiac arrest patient", section on 'Temperature control'.)

Methods of enhancing the elimination of poisons, such as gastric lavage, activated charcoal, whole bowel irrigation, and hemodialysis, may be helpful for selected patients (see "Gastrointestinal decontamination of the poisoned patient"). Circumstances may suggest the presence of concealed drug packages in the GI tract or vagina. (See "Internal concealment of drugs of abuse (body packing)".)

CESSATION OF RESUSCITATIVE EFFORTS

Unlike most cardiac arrest patients, prolonged resuscitation efforts are not necessarily futile for poisoned patients [27-29]. Neurologically intact survival after more than 60 minutes of pulseless arrest and external chest compressions is possible [30-32]. Prolonged resuscitative efforts including extracorporeal circulatory assistance may be warranted for previously healthy patients with witnessed cardiac arrest following drug overdose [14,33]. Even if neurologic recovery is not possible, selected patients poisoned with neurotoxins such as methanol, carbon monoxide, and cyanide can serve as organ donors.

ADDITIONAL RESOURCES

Regional poison centers — Regional poison centers in the United States are available at all times for consultation on patients with known or suspected poisoning, and who may be critically ill, require admission, or have clinical pictures that are unclear (1-800-222-1222). In addition, some hospitals have medical toxicologists available for bedside consultation. Whenever available, these are invaluable resources to help in the diagnosis and management of ingestions or overdoses. Contact information for poison centers around the world is provided separately. (See "Society guideline links: Regional poison centers".)

Society guideline links — Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: General measures for acute poisoning treatment" and "Society guideline links: Treatment of acute poisoning caused by specific agents other than drugs of abuse".)

SUMMARY AND RECOMMENDATIONS

- **Protection of clinicians and emergency department (ED)** – While resuscitating a critically ill adult with an unknown poisoning, the clinician must preserve the operational capacity of the emergency health care system and ensure the safety of health care workers. In addition to multiple casualty scenarios, even a single patient contaminated or potentially contaminated with a highly potent toxin can disable an entire ED. (See 'First priorities' above and 'Initial data acquisition' above.)

- **Examination and identification of toxin** – During the resuscitation, characteristic combinations of selected findings (so called "toxidromes") (table 1) should be rapidly sought, including: vital signs; level of alertness; pupil size and position; mucous membrane moisture and secretions; skin temperature and moisture; presence or absence of bowel sounds; and motor tone. Diagnostic tests to be obtained as part of the initial evaluation are described in the text. (See 'Rapid first look: Examination, monitoring, and testing' above.)

- **Management** – A systematic approach is essential to the resuscitation of the critically ill adult with an unknown overdose and can be adapted from the usual "ABC" (airway, breathing, circulation) scheme. The steps are organized according to the issues that pose the most immediate life threats: airway, breathing, circulation, disability (neurologic stabilization), and exposure and elimination. Problems are managed immediately and concurrently as identified. Do not solely use a Glasgow Coma Scale score ≤8 to identify the need for tracheal intubation in an unresponsive poisoned patient who is protecting their airway, maintaining adequate oxygen saturation, hemodynamically stable, and expected to not deteriorate based on the suspected ingestion. (See 'Systematic evaluation: The "ABCDE" approach' above.)

- **Prolonged resuscitation** – Unlike most cardiac arrest patients, prolonged resuscitation efforts are not necessarily futile for poisoned patients. Neurologically intact survival after more than 60 minutes of pulseless arrest and external chest compressions is possible. (See 'Cessation of resuscitative efforts' above.)

ACKNOWLEDGMENT

The UpToDate editorial staff acknowledges Stephen J Traub, MD, former section editor of the toxicology program, for 20 years of dedicated service.

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. Tanz LJ, Dinwiddie AT, Mattson CL, et al. Drug Overdose Deaths Among Persons Aged 10-19 Years - United States, July 2019-December 2021. MMWR Morb Mortal Wkly Rep 2022; 71:1576.
2. Spencer MR, Miniño AM, Warner M. Drug Overdose Deaths in the United States, 2001-2021. NCHS Data Brief 2022; :1.
3. Albertson TE, Dawson A, de Latorre F, et al. TOX-ACLS: toxicologic-oriented advanced cardiac life support. Ann Emerg Med 2001; 37:S78.
4. Panchal AR, Bartos JA, Cabañas JG, et al. Part 3: Adult Basic and Advanced Life Support: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2020; 142:S366.
5. Traub SJ, Hoffman RS, Nelson LS. Body packing--the internal concealment of illicit drugs. N Engl J Med 2003; 349:2519.
6. Del Pozo B, Sightes E, Kang S, et al. Can touch this: training to correct police officer beliefs about overdose from incidental contact with fentanyl. Health Justice 2021; 9:34.
7. Moss MJ, Warrick BJ, Nelson LS, et al. ACMT and AACT Position Statement: Preventing Occupational Fentanyl and Fentanyl Analog Exposure to Emergency Responders. J Med Toxicol 2017; 13:347.
8. De Groot R, Van Zoelen GA, Leenders MEC, et al. Is secondary chemical exposure of hospital personnel of clinical importance? Clin Toxicol (Phila) 2021; 59:269.
9. Sivilotti ML. Flumazenil, naloxone and the 'coma cocktail'. Br J Clin Pharmacol 2016; 81:428.
10. Nanah A, Abdeljaleel F, Matsubara JK, Garcia MVF. Outcomes and Practices of Endotracheal Intubation Using the Glasgow Coma Scale in Acute Non-Traumatic Poisoning: A Systematic Review and Meta-Analysis of Proportions. J Intensive Care Med 2025; 40:1143.
11. Freund Y, Viglino D, Cachanado M, et al. Effect of Noninvasive Airway Management of Comatose Patients With Acute Poisoning: A Randomized Clinical Trial. JAMA 2023; 330:2267.
12. Adnet F, Borron SW, Finot MA, et al. Intubation difficulty in poisoned patients: association with initial Glasgow Coma Scale score. Acad Emerg Med 1998; 5:123.
13. Skoog CA, Engebretsen KM. Are vasopressors useful in toxin-induced cardiogenic shock? Clin Toxicol (Phila) 2017; 55:285.
14. de Lange DW, Sikma MA, Meulenbelt J. Extracorporeal membrane oxygenation in the treatment of poisoned patients. Clin Toxicol (Phila) 2013; 51:385.
15. Upchurch C, Blumenberg A, Brodie D, et al. Extracorporeal membrane oxygenation use in poisoning: a narrative review with clinical recommendations. Clin Toxicol (Phila) 2021; 59:877.
16. Gosselin S, Hoegberg LC, Hoffman RS, et al. Evidence-based recommendations on the use of intravenous lipid emulsion therapy in poisoning(). Clin Toxicol (Phila) 2016; 54:899.
17. Levine M, Hoffman RS, Lavergne V, et al. Systematic review of the effect of intravenous lipid emulsion therapy for non-local anesthetics toxicity. Clin Toxicol (Phila) 2016; 54:194.
18. Warrick BJ, Tataru AP, Smolinske S. A systematic analysis of methylene blue for drug-induced shock. Clin Toxicol (Phila) 2016; 54:547.
19. Chan A, Isbister GK, Kirkpatrick CM, Dufful SB. Drug-induced QT prolongation and torsades de pointes: evaluation of a QT nomogram. QJM 2007; 100:609.
20. Burns MJ, Dickson EW, Sivilotti ML, Cuenoud H. Phentolamine reduces myocardial injury and mortality in a rat model of phenylpropanolamine poisoning. J Toxicol Clin Toxicol 2001; 39:129.

21. Wood DM, Dargan PI, Hoffman RS. Management of cocaine-induced cardiac arrhythmias due to cardiac ion channel dysfunction. Clin Toxicol (Phila) 2009; 47:14.
22. Wu S, Pearl-Davis MS, Manini AF, Hoffman RS. Use of antipsychotics to treat cocaine toxicity? Acad Emerg Med 2008; 15:105; author reply 106.
23. Hoffman RS. Cocaine and beta-blockers: should the controversy continue? Ann Emerg Med 2008; 51:127.
24. Hoffman RS, Goldfrank LR. The poisoned patient with altered consciousness. Controversies in the use of a 'coma cocktail'. JAMA 1995; 274:562.
25. Jackson R, Teece S. Best evidence topic report. Oral or intravenous thiamine in the emergency department. Emerg Med J 2004; 21:501.
26. Fletcher ML, Sarangarm P, Nash J, et al. A systematic review of second line therapies in toxic seizures. Clin Toxicol (Phila) 2021; 59:451.
27. Morrison LJ, Verbeek PR, Zhan C, et al. Validation of a universal prehospital termination of resuscitation clinical prediction rule for advanced and basic life support providers. Resuscitation 2009; 80:324.
28. Kellermann AL, Hackman BB, Somes G. Predicting the outcome of unsuccessful prehospital advanced cardiac life support. JAMA 1993; 270:1433.
29. Bonnin MJ, Pepe PE, Kimball KT, Clark PS Jr. Distinct criteria for termination of resuscitation in the out-of-hospital setting. JAMA 1993; 270:1457.
30. Ramsay ID. Survival after imipramine poisoning. Lancet 1967; 2:1308.
31. Southall DP, Kilpatrick SM. Imipramine poisoning: survival of a child after prolonged cardiac massage. Br Med J 1974; 4:508.
32. Orr DA, Bramble MG. Tricyclic antidepressant poisoning and prolonged external cardiac massage during asystole. Br Med J (Clin Res Ed) 1981; 283:1107.
33. International Liaison Committee on Resuscitation. 2005 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science with Treatment Recommendations. Part 4: Advanced life support. Resuscitation 2005; 67:213.

Topic 13850 Version 35.0

GRAPHICS

Table 1: **Common poisoning syndromes (toxidromes)**

Toxidrome	Mental status	Vital signs	Skin	Pupils	Other manifestations	Examples of causative agents
Excitatory						
Sympathomimetic	▪ Hypervigilance ▪ Agitation (can be violent) ▪ Hyperactive delirium ▪ Hallucinations ▪ Paranoia	**T:** Increased **HR:** Increased **RR:** Increased **BP:** Increased	Wet	Dilated	▪ Seizures ▪ Widened pulse pressure	▪ Amphetamines ▪ Cocaine ▪ Cathinones ▪ Ephedrine ▪ Phenylpropanolamine ▪ Pseudoephedrine
Anticholinergic	▪ Hypervigilance ▪ Agitation (usually easily controlled) ▪ Hyperactive delirium ▪ Hallucinations (picking at objects in air) ▪ Mumbling speech (described as "mouth full of marbles")	**T:** Increased **HR:** Increased (but may be normal in early poisoning) **RR:** Increased **BP:** Increased or normal	Dry and flushed	Dilated	▪ Dry mucous membranes ▪ Decreased bowel sounds ▪ Urinary retention ▪ Choreoathetosis ▪ Seizures (rare)	▪ Diphenhydramine (and other antihistamines) ▪ Atropine and similar agents (hyoscyamine, dicyclomine, scopolamine, and naturally occurring belladonna alkaloids [eg, jimson weed]) ▪ Tricyclic antidepressants ▪ Cyclobenzaprine ▪ Orphenadrine ▪ Phenothiazines
Hallucinogenic	▪ Hallucinations ▪ Perceptual distortions (typically visual) ▪ Depersonalization ▪ Synesthesia ▪ Occasional agitation (with or without delirium)	**T:** Increased or normal **HR:** Increased or normal **RR:** Increased or normal **BP:** Increased or normal	Variable	Dilated (usually)	▪ Nystagmus (phencyclidine, ketamine) ▪ Tachycardia, hypertension, agitated delirium (designer phenethylamines)	▪ Designer phenethylamines and tryptamines (eg, MDMA ["ecstasy"], MDEA) ▪ Ketamine and methoxetamine ▪ LSD and psilocybin ▪ Phencyclidine ▪ Mescaline
Serotonin syndrome (serotonin toxicity)	▪ Agitation ▪ Hyperactive delirium ▪ Confusion ▪ Awake and unresponsive	**T:** Increased **HR:** Increased **RR:** Increased **BP:** Increased	Wet, flushed, or normal	Dilated	▪ Tremor, hyperreflexia, clonus (typically in lower extremities) ▪ Roving eye movements (ocular clonus) ▪ Diarrhea	▪ MAOIs ▪ Tricyclic antidepressants ▪ SSRIs and SNRIs ▪ Dextromethorphan ▪ Meperidine ▪ Refer to UpToDate content on combinations of agents that can cause serotonin syndrome
Inhibitory						

Opioid	▪ Sedation ▪ Coma	**T:** Decreased or normal **HR:** Decreased or normal **RR:** Decreased or apneic **BP:** Decreased or normal	Variable	Constricted (may be pinpoint)	▪ Noncardiogenic pulmonary edema ▪ Needle marks ▪ Can develop hypotension	▪ Opioids (eg, fentanyl and analogues, heroin, morphine, methadone, oxycodone, hydromorphone) ▪ Diphenoxylate ▪ Loperamide
Sedative-hypnotic	▪ Sedation ▪ Confusion ▪ Stupor ▪ Coma	**T:** Decreased or normal **HR:** Decreased or normal **RR:** Decreased, apneic, or normal **BP:** Decreased or normal	Variable	Variable	▪ Nystagmus ▪ Barbiturates can cause respiratory depression or apnea ▪ In most cases, isolated benzodiazepine ingestions do not cause respiratory depression ▪ Cyclical coma and myoclonic encephalopathy (carisoprodol, meprobamate, glutethimide)	▪ Benzodiazepines ▪ Barbiturates ▪ Ethanol and other alcohols ▪ Gabapentin and pregabalin ▪ Zolpidem ▪ Carisoprodol ▪ Glutethimide ▪ Meprobamate
Cholinergic	▪ Sedation ▪ Confusion ▪ Stupor ▪ Coma	**T:** Normal **HR:** Low (may be increased in early poisoning) **RR:** Decreased or increased **BP:** Decreased or normal	Wet	Constricted	▪ Seizures (typically occur early) ▪ Salivation ▪ Urinary and fecal incontinence ▪ Vomiting, diarrhea, abdominal cramps ▪ Bronchorrhea and bronchoconstriction ▪ Muscle fasciculations and paralysis ▪ Weakness	▪ Organophosphate and carbamate insecticides ▪ Nerve agents (eg, VX, tabun, sarin, soman, and Novichok) ▪ Nicotine ▪ Physostigmine ▪ Rivastigmine ▪ Bethanechol ▪ Pilocarpine ▪ Urecholine

BP: blood pressure; HR: heart rate; MAOIs: monoamine oxidase inhibitors; MDEA: 3,4-methylenedioxy-N-ethylamphetamine; MDMA: 3,4-methylenedioxymethamphetamine; RR: respiratory rate; SNRIs: serotonin-nonspecific reuptake inhibitors; SSRIs: serotonin-specific reuptake inhibitors; T: temperature; VX: venomous agent X.

Graphic 71268 Version 25.0

Table 2: **Drug- and toxin-induced temperature abnormalities**

Hyperthermia	Hypothermia

Increased heat production	▪ Opioids
	▪ Sedative-hypnotics
	  • Benzodiazepines
	  • Barbiturates
	  • Alcohols
	▪ Sympatholytics
	  • Beta blockers
	  • ClonidineText
	  • Alpha-adrenergic antagonists
	▪ Hypoglycemic agents
	▪ Antipsychotics
	▪ General anesthetic agents
	▪ Carbon monoxide
	▪ Drugs that cause flaccid coma

▪ Muscular hyperactivity/rigidity
  • Sympathomimetics
  ◦ Cocaine
  ◦ Amphetamines
  ◦ Phenylpropanolamine
  ◦ Ephedrine
  ◦ Cathinones
  • Anticholinergics
  • Drug withdrawal states
  • Lithium
  • Central hallucinogens
  ◦ Phencyclidine
  ◦ Lysergic acid diethylamide (LSD)
  ◦ Designer amphetamines (MDMA, MDEA)
  ◦ Synthetic cannabinoids
  • Drugs causing recurrent seizures
  ◦ Isoniazid
  ◦ Theophylline
  ◦ Phenethylamines
  • Strychnine
  • Neuroleptic malignant syndrome
  • Serotonin syndrome
  • MAO inhibitors
  • Malignant hyperthermia	

Impaired heat dissipation	

▪ Impaired sweating
  • Anticholinergic agents
  ◦ Antihistamines
  ◦ Phenothiazines
  ◦ Tricyclic antidepressants	

Increased metabolic rate	

▪ Uncoupled oxidative phosphorylation
  • Salicylates
  • Dinitrophenol, pentachlorophenol
▪ Thyroid hormone	

MAO: monoamine oxidase; MDEA: 3,4-methylenedioxyethylamphetamine; MDMA: 3,4-methylenedioxymethamphetamine.

Graphic 55439 Version 10.0

Table 3: **Drug- and toxin-induced changes in blood pressure and pulse**

Hypertension with tachycardia	Hypertension with bradycardia	Hypotension with tachycardia	Hypotension with bradycardia
Sympathomimetics	Alpha-adrenergic agonists	Beta-adrenergic agonists	Beta-blockers
Amphetamines	Phenylpropanolamine	Theophylline	Calcium-channel blockers
Cocaine	Phenylephrine	Albuterol	Cardiac glycosides
Ephedrine	Phentermine	Isoproterenol	Digoxin
Pseudoephedrine	Ergot alkaloids	Terbutaline	Digitalis purpurea
Theophylline	Sumatriptan	Caffeine	Oleander
Caffeine	Clonidine (early)	Disulfiram reaction (late)	Red squill
Methylphenidate	Guanfacine	Toxic alcohols	Bufotenin
Cathinones	Imidazolines	Isopropyl alcohol	Clonidine
Anticholinergics	Tetrahydrozoline	Carbon monoxide	Alpha-methyldopa
Antihistamines	Oxymetazoline	Alpha-adrenergic antagonists	Cyanide
TCAs (early)	Cholinergic agents	Phenothiazines	Carbon monoxide (late)
Phenothiazines (some)	Organophosphates	TCAs	Opioids
Antiparkinson agents	Carbamates	Hydralazine	Sedative-hypnotics
Muscle relaxants	Steroid hormones	Heavy metals (acute)	Barbiturates
Clozapine	Glucocorticoids	Iron	Benzodiazepines
Central hallucinogens	Mineralocorticoids	Arsenic	Cholinergics
Designer amphetamines	Estrogen	Colchicine	Organophosphates
Lysergic acid diethylamide (LSD)	Progesterone	Nitrates	Carbamates
Phencyclidine (PCP)	Androgens	Sodium nitroprusside	Antiarrhythmics
Synthetic cannabinoids	Yohimbine		
Envenomations	Heavy metals		
Black widow spider bite	Lead		
Scorpion stings	Disulfiram reaction (early)		
Drug withdrawal states			
MAOIs (foods with tyramine)			
Nicotine			
Cholinergic agents (sometimes)			
Organophosphates			
Carbamates			
Thyroid hormone			

Graphic 51200 Version 8.0

Table 4: **Drugs and toxins associated with respiratory dysfunction**

Tachypnea/hyperventilation	Bradypnea/hypoventilation
Sympathomimetics	CNS depressants
Amphetamines	Opioids
Cocaine	Sedative-hypnotics
Caffeine	Alcohols
Theophylline	Antidepressants
Nicotine	Antipsychotics
Cathinones	Sympatholytics
Central hallucinogens	Volatile inhalants (solvents)
Lysergic acid diethylamide (LSD)	Cholinergics
Phencyclidine	Muscle relaxants
Designer amphetamines	Antiseizure medications
Synthetic cannabinoids	Respiratory muscle failure
Anticholinergics	Botulism
Drug withdrawal states	Carbamates
Salicylates	Neurotoxic snake envenomation
Dinitrophenol, pentacholorphenol	Neuromuscular blocking agents
Drug-associated hepatic failure	Organophosphates
Acetaminophen	Paralytic shellfish poisoning (saxitoxin)
Amanita mushrooms	Puffer fish poisoning (tetrodotoxin)
Cellular asphyxiants	Strychnine
Carbon monoxide	Tetanus
Cyanide	
Hydrogen sulfide	
Methemoglobinemia	
Toxins that induce pulmonary edema	
Opioids	
Pulmonary irritants	
Drugs that induce metabolic acidosis (respiratory compensation)	
Methanol	
Ethylene glycol	
Alcoholic ketoacidosis	
Iron	
Isoniazid	

Graphic 64012 Version 6.0

Table 5: **Glasgow Coma Scale (GCS)**

	Score
Eye opening	
Spontaneous	4
Response to verbal command	3
Response to pain	2
No eye opening	1
Best verbal response	
Oriented	5
Confused	4
Inappropriate words	3
Incomprehensible sounds	2
No verbal response	1
Best motor response	
Obeys commands	6
Localizing response to pain	5
Withdrawal response to pain	4
Flexion to pain	3
Extension to pain	2
No motor response	1
Total	

The GCS is scored between 3 and 15, 3 being the worst and 15 the best. It is composed of three parameters: best eye response (E), best verbal response (V), and best motor response (M). The components of the GCS should be recorded individually; for example, E2V3M4 results in a GCS score of 9.

In the setting of head trauma, a GCS score of 8 or less measured on admission represents severe traumatic brain injury (TBI).

Traditionally, a GCS score of 9 through 12 has represented moderate TBI, and a GCS score of 13 through 15 mild TBI. However, the recognition that more than one-third of patients with TBI and a GCS score of 13 have potentially life-threatening intracranial lesions has led to a reevaluation of this classification. While a revised classification has not been widely adopted, a GCS score of 9 through 13 likely best represents the TBI population at moderate risk for death or long-term disability.[1]

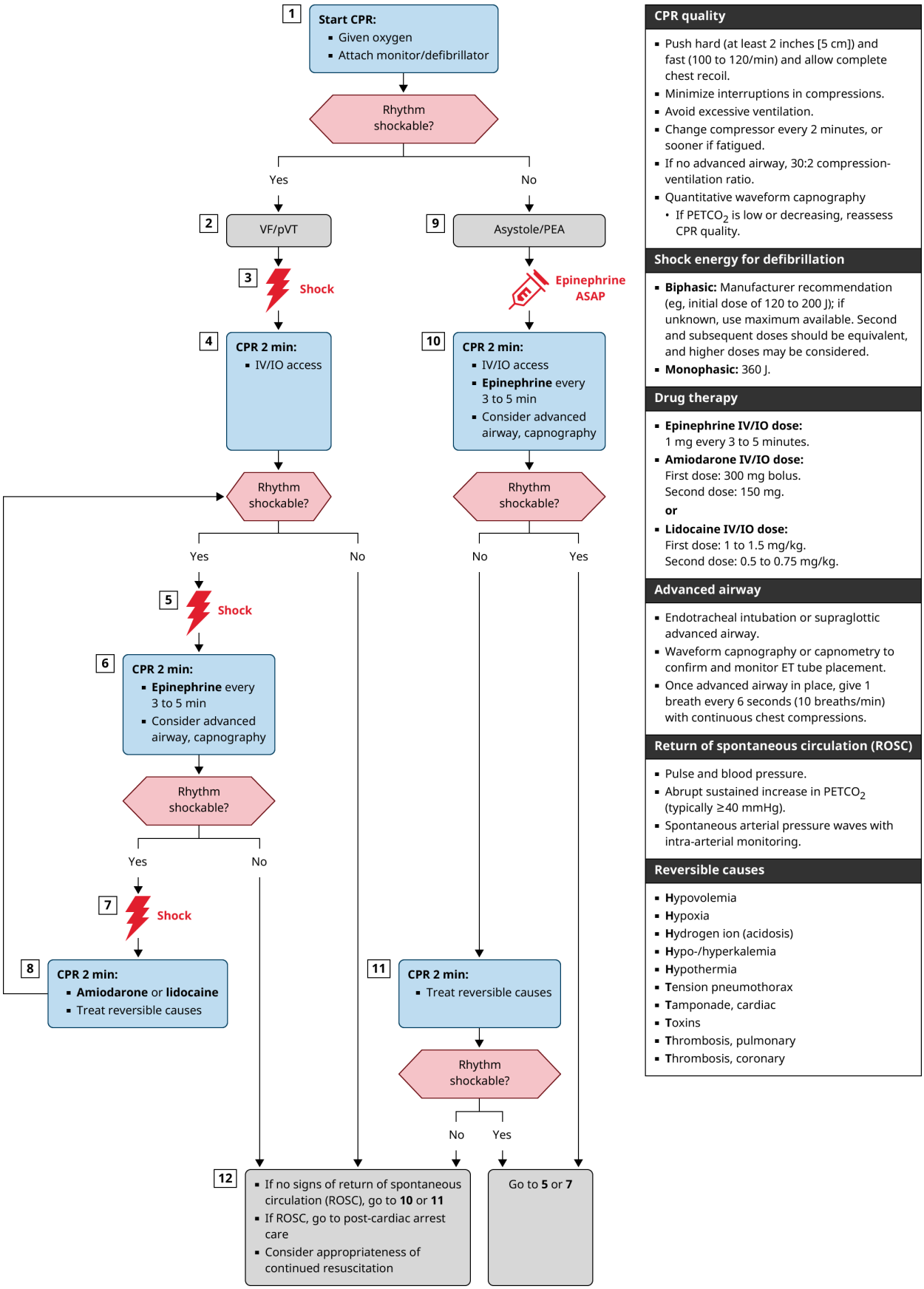
Reference:

1. *Godoy DA, Aguilera S, Rabinstein AA. Potentially severe (moderate) traumatic brain injury: A new categorization proposal. Crit Care Med 2020; 48:1851.*

Reproduced with permission from: Teasdale G, Jennett B. Assessment of coma and impaired consciousness: A practical scale. Lancet 1974; 2:81. Copyright © by the Lancet Ltd. 1974.

Graphic 81854 Version 11.0

Algorithm 1: **Adult cardiac arrest algorithm**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Adult Cardiac Arrest Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ET: endotracheal; IO: intraosseous; IV: intravenous; J: joules; PEA: pulseless electrical activity; $PETCO_2$: peak end-tidal carbon dioxide; pVT: pulseless ventricular tachycardia; ROSC: return of spontaneous circulation; VF: ventricular fibrillation.

Reprinted with permission. Highlights of the 2020 American Heart Association Guidelines for CPR and ECC. Copyright © 2020 American Heart Association, Inc.

Graphic 129983 Version 14.0

Table 6: **Successful use of ECMO in respiratory or cardiac failure due to poisonings**

Ingestions successfully managed with ECMO
Calcium channel blockers (eg, verapamil, diltiazem, amlodipine)
Beta-blockers (eg, propranolol, metoprolol, atenolol, acebutolol)
Antiarrhythmics (eg, flecainide, quinidine, propafenone, amiodarone)
Other cardiovascular medications (eg, lisinopril, clonidine, digoxin)
Antidepressants (eg, amitriptyline, doxepin, venlafaxine, bupropion, trazodone, citalopram)
Antipsychotics (eg, fluphenazine, quetiapine, risperidone)
Sedative/hypnotics (eg, zolpidem, meprobamate)
Stimulants (eg, cocaine, methamphetamine, MDMA, caffeine)
Anticonvulsants (eg, carbamazepine, phenytoin/fosphenytoin, gabapentin, lamotrigine)
Antineoplastics (eg, 5-fluorouracil, bleomycin)
Local anesthetics (eg, lidocaine, mepivacaine)
Toxic alcohol (eg, ethylene glycol, methanol)
Analgesics (eg, methadone, tramadol, ibuprofen, aspirin)
Antihistamines (eg, diphenhydramine, hydroxyzine)
Hydrocarbons (ie, pneumonitis)
Herbicides/pesticides (eg, paraquat, diquat, glyphosate, aluminum phosphide)
Cellular asphyxiants (eg, carbon monoxide, sodium azide, hydrogen sulfide)
Alkaloids (aconitine, colchicine, taxine/yew)
Chloroquine
Loperamide
Metformin

This is not an exhaustive list of ingestions or exposures. ECMO can be used as a temporizing measure for a poisoning that causes circulatory and/or respiratory collapse until the toxicant is metabolized and organ function has recovered.

ECMO: extracorporeal membrane oxygenation; MDMA: 3,4-methylenedioxymethamphetamine (ie, "ecstasy").

References:

1. *Upchurch C, Blumenberg A, Brodie D, et al. Extracorporeal membrane oxygenation use in poisoning: a narrative review with clinical recommendations. Clin Toxicol (Phila) 2021; 59:877.*
2. *Voicu S, M'Rad A, Malissin I, et al. Extracorporeal life support in cardiotoxicant poisoning-A narrative review. Basic Clin Pharmacol Toxicol 2023; 132:5.*
3. *Pozzi M, Buzzi R, Hayek A, et al. Veno-arterial extracorporeal membrane oxygenation for drug intoxications: A single center, 14-year experience. J Card Surg 2022; 37:1512.*
4. *Duburcq T, Goutay J, Preau S, et al. Venoarterial extracorporeal membrane oxygenation in severe drug intoxication: A retrospective comparison of survivors and nonsurvivors. ASAIO J 2022; 68:907.*
5. *Orlando A, Sciutti F, Colombo CN, et al. Ethylene glycol poisoning requiring veno-arterial ECMO: A case report. Perfusion 2022.*

Graphic 140779 Version 1.0

Table 7: **Empirical dosing of emergency antidotes to adult patients in extremis***

Suspected toxin (or similar agent)	Antidote
Cyanide	5 g hydroxocobalamin IV, repeat x 1 (alternative is nitrite/thiosulfate kit)
Methemoglobinemia	2 mL/kg 2 percent methylene blue IV over 5 minutes
Digoxin	2 to 5 vials digoxin antibody (Fab) fragments IV
Calcium channel blocker	20 mL 10 percent calcium chloride IV, repeat x 2; 0.5 to 1 units/kg/hour insulin (and dextrose) IV
Local anesthetic	1.5 mL/kg of 20 percent intralipid bolus IV, may repeat several times (alternatively, after bolus infuse 0.25 mL/kg per minute for 30 minutes; maximum dose 10 mL/kg)
Sodium channel blockade	100 mL of 7.5 percent to 8.4 percent sodium bicarbonate IV, repeat x 2
Cholinergic agent	2 mg atropine IV , doubled every 3 minutes as needed
Isoniazid	5 g pyridoxine over 10 minutes or until seizures stop; remainder over 4 hours
Chloroquine	2 mg/kg diazepam IV over 30 minutes

* Additional details about antidotes and further treatment of the poisons listed in this table can be found in the specific reviews discussing these poisonings.

Graphic 66622 Version 5.0

Contributor Disclosures

Marco L A Sivilotti, MD, MSc, FRCPC, FACMT, FAACT No relevant financial relationship(s) with ineligible companies to disclose. **Evan Schwarz, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Michael Ganetsky, MD** No relevant financial relationship(s) with ineligible companies to disclose.

Contributor disclosures are reviewed for conflicts of interest by the editorial group. When found, these are addressed by vetting through a multi-level review process, and through requirements for references to be provided to support the content. Appropriately referenced content is required of all authors and must conform to UpToDate standards of evidence.

Conflict of interest policy

→